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Data Science

Assignment 4

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Introduction

This report presents the implementation and analysis of our deep learning models developed as part of Assignment 4 for the Data Science course. The project is structured around three distinct tasks, each focusing on a unique data domain and neural network architecture, collectively equipping us with a comprehensive deep learning workflow.

The first section (**Task 1**) focuses on feature engineering and tabular data regression. Using the California Housing dataset, we train a **Multi-Layer Perceptron (MLP)** to predict median house prices. This task involves a complete end-to-end pipeline, including exploratory data analysis, data scaling, baseline modeling, and the manual creation of domain-specific features to evaluate their impact on the model's generalization and prediction accuracy.

The second section (**Task 2**) introduces computer vision by building a **Convolutional Neural Network (CNN)** for image classification on the Fruits and Vegetables dataset. Beyond achieving high classification accuracy, this section emphasizes the mathematical foundations of convolutional layers and rigorous preprocessing to prevent data leakage. We conduct a deep robustness analysis to determine whether the model learns actual morphological shapes or merely memorizes colors, and we utilize **Grad-CAM** to visually explain the model's spatial attention and decision-making process.

The third section (**Task 3**) implements sequential modeling to solve a **Knowledge Tracing** problem within educational data mining. Using the ASSISTments2017 dataset, we develop a **Recurrent Neural Network (RNN)** to analyze chronological student interactions and predict future correctness. This layer explores sequence windowing, temporal feature engineering, and the critical handling of varying sequence lengths, culminating in an advanced **LSTM-based** implementation to better capture long-term learning dependencies.

Task 1: MLP

2.1 Dataset Loading

2.2 Exploratory Data Analysis (EDA)

2.3 Data Preparation

2.4 Baseline MLP Model

2.5 Model Training

2.6 Feature Engineering

2.7 Enhanced MLP Model

2.8 Model Evaluation and Comparison

2.9 Error Analysis

2.10 Conclusion and Future Work

Task 2: CNN

Task 3: RNN

4.1 Introduction

4.2 Dataset Description

4.3 Preprocessing and Feature Engineering

4.4 Exploratory Data Analysis (EDA)

4.5 Model Architecture

4.6 Model Training

4.7 Evaluation and Visualization

4.8 Implementing an LSTM-Based Model

Questions

Conclusion